

Unsupervised Learning Project: A Reproducible Experimental Study of Clustering

SUSANA NASCIMENTO (SNT@FCT.UNL.PT)

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Objectives & research questions

- **Formulate** and **motivate** a clustering problem without class labels (RQ1).
- **Design a reproducible pipeline** for mixed-type tabular data (RQ2).
- **Compare models**: k -means baseline + $\geq$ one additional clustering family (RQ3).
- Define “*what to measure*”: internal validity + stability + problem-relevant proxies (RQ3/RQ5).
- **Interpret clusters in context**: profiles, limitations, and decision-relevant recommendations (RQ4).
- **Deliverable**: 6–8 page academic report + fully reproducible repository (run_all, experiments log, environment specification).

Hotel Booking Demand — dataset in context

Dataset Description

- Real-world hotel Property Management System (PMS) data (Portugal).
- Two hotels: Resort (Algarve) and City (Lisbon).
- Unit of analysis: one row = one booking (arrived or cancelled).
- Arrivals span: 1 Jul 2015 → 31 Aug 2017.
- Identifiers removed/anonymised; engineered variables documented in the reference paper.
- Feature values are generally taken as recorded on the day prior to scheduled arrival (from the PMS change log when available), a design choice intended to reduce future-information leakage in cancellation prediction (not a substitute for leakage control in clustering).

119,390

Bookings

2

Hotels

32

Attributes

2015–2017

Arrival window

Bookings by hotel (course release v1)



Source: António, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. Data in Brief, 22, 41–49. <https://doi.org/10.1016/j.dib.2018.11.126> (CC BY 4.0).

Abbreviations: PMS = Property Management System; ADR = Average Daily Rate; TA/TO = Travel Agents / Tour Operators.

Problem Formulation for Unsupervised Analysis

Illustrative RQ1 Examples

Your report must define a concrete RQ1 and justify why clustering is appropriate.

Demand planning / seasonality

Research question: Do distinct lead-time × seasonality profiles emerge (e.g., last-minute vs early planners), and how stable are they?

Decision relevance: Supports staffing and capacity planning.

Channel strategy

Research question: Are there coherent “booking profiles” associated with the booking distribution channels and market segments (*self-booked vs intermediary-booked vs business-account-booked*)?

Decision relevance: Supports marketing spend and channel management.

Stay patterns & product

Research question: Do weekend vs weekday stays form distinct segments, and how do they relate to room types and special requests?

Decision relevance: Supports product design and operational planning.

Value & policy

Research question: Are there high-value vs low-value segments with distinct booking-change behaviour and deposit choices?

Decision relevance: Supports pricing and policy design.

Methodological note: Define the **index time** (the time at which features are measured/available) and exclude outcome/post-event variables from the clustering inputs (to prevent information leakage).

RQ1 Examples are illustrative; teams must formulate and justify their own RQ1 and feature set under the project brief constraints.

Variables and structure

Mixed-type tabular data (numeric + categorical + date-like fields)

Booking timing

Lead time
Arrival (year/month/week/day)
Seasonality proxies

Stay & party

Nights (week/weekend)
Adults/children/babies
Room type, requests

Channel & customer

Distribution channel
Market segment, customer type
Country (high-cardinality)

Value & policy

(ADR) Average Daily Rate/pricing
Deposit type, waiting list
Booking changes

High-cardinality / ID-like fields

- country ≈ 177 categories
- agent ID ≈ 333 values
- company ID ≈ 352 values

Implication: define a responsible handling strategy (drop / group rare levels / thresholding / justified encoding).

Reminder: outcome/post-event fields are retained only for post-hoc profiling (not as clustering inputs).

Full variable dictionary + engineering notes: Table 1 in António *et al.* (2019) [doi:10.1016/j.dib.2018.11.126](https://doi.org/10.1016/j.dib.2018.11.126)

Dataset & feature governance

Hotel Booking Demand — at a glance

A real-world, mixed-type dataset of hotel booking records. Your goal is to discover coherent “booking profiles” (groups) without labels.

Booking timing

Lead time, arrival info, seasonality

Stay & party

Nights, guests, room types, requests

Channel & customer

Market segment, channel, country

Value & policy

ADR/pricing, deposits, booking options

Note: outcome/status fields exist for profiling (e.g., cancellation/status) — they must be excluded from clustering inputs.

Non-negotiable constraints

- Leakage control: exclude outcome/post-event variables from clustering inputs (min: is_canceled, reservation_status).
- Remove ID-like / very high-cardinality fields unless justified and responsibly encoded.
- Predefine and justify hyperparameter ranges; unbounded searches are not permitted.
- Distance/metric governance: state representation + implied distance; compute indices consistently.

Data quality and modelling implications

What you should report (Task 1.1)

- Missingness by type (numeric vs categorical).
- Rare categories and high-cardinality fields.
- Outliers / heavy tails (justify any filtering).
- Final feature set + implied distance/similarity.
- Recompute all descriptive statistics on the provided course release (*do not copy tables/figures from the reference paper*).

Examples of extreme values (course release v1)

Variable	Observed range
ADR	−6.38 to 5400
Lead time	0 to 737 days
Adults	0 to 55
Week nights	0 to 50
Waiting list	0 to 391 days

Ranges and caveats are consistent with the dataset reference paper (Tables 4–7) and the project brief requirements.

Interpretation caveats

- In the reference paper, “NULL” values in agent/company are described as “not applicable”, not as missing data.
- Some attributes may only be reliably known at check-in (e.g., nationality), so distributions can differ between cancelled vs non-cancelled bookings.
- Choose scaling/encoding with these properties in mind; document trade-offs explicitly.

Methodological implication

Robustness is part of the deliverable: assess sensitivity to reasonable preprocessing variants (e.g., StandardScaler vs RobustScaler; with vs without PCA/SVD) and report stability across seeds.

Dataset Integrity, Feature Selection and Leakage Control

- You will work with the Hotel Booking Demand dataset provided as “**course release v1**” (a fixed snapshot).
- Verify you are using the exact same data by checking the SHA-256 in **SHA256SUMS.txt** and cite the version + hash in your report.
- Use **DATASET_MANIFEST.yml** for the dataset documentation (source, reference, unit of analysis, and license/terms).
- Follow **column_roles.csv** for feature governance: do not use outcome/post-event variables (e.g., `is_canceled`, `reservation_status`) to form clusters, and treat ID-like/high-cardinality fields (e.g., `agent/company`) cautiously.
- If a method is computationally too heavy on the full dataset, a reproducible **n=30,000** subsample (seeded indices) is provided—report and discuss any impact if you use it.
- Do not commit the raw CSV to version control (i.e. any repo system); provide scripts/instructions to obtain it and reproduce preprocessing end-to-end.

Milestones and expected artefacts

The project is structured around checkpoints that progressively enforce methodological discipline and reproducibility.

Date	Checkpoint (what you must have)	Focus
Fri 20 Mar 2026	Pitch: problem statement; dataset doc (draft); success criteria; risks/ethics; repo scaffold.	Feasibility & governance
Fri 24 Apr 2026	Mid-term progress: completed EDA; ≥ 2 baselines with indices; sensitivity plan; experiment log.	Technical progress
Fri 5 Jun 2026 (23:59)	Final submission: 6–8 page academic report + fully reproducible repository (<code>run_all</code>).	Complete study

Internal milestones (recommended):

- **Wed 13 May** — Core protocol freeze (parameter ranges fixed; seeds locked).
- **Wed 20 May** — Reproducibility freeze (environment final; `run_all` end-to-end).
- **Wed 27 May** — Final readiness review (short rehearsal; Q&A preparation).

Dates and checkpoint requirements follow the project brief.

Core workflow & mandatory checkpoints

Core (mandatory) tasks — up to 13 points

- **Task 1:** Preprocessing pipeline + k -means baseline + iK-means.
- **Task 2:** Add one distinct family (hierarchical / GMM / DBSCAN/OPTICS).
- **Task 3:** Evaluation protocol: internal indices + diagnostics + stability (≥ 5 seeds/resamples) + 1 controlled preprocessing variant.
- **Task 4:** 6–8 page report + fully reproducible repository (run_all, experiments.csv, environment).

Mandatory deliverables (checkpoints)

Fri 20 Mar (after Lab P4)	Pitch: problem statement; dataset documentation (draft); success criteria; risks/ethics; repo scaffold
Fri 24 Apr (after Lab P8)	Mid-term progress: EDA; ≥ 2 baselines + indices; sensitivity plan; experiment logging
Fri 5 Jun 2026 23:59	Final submission: academic report + reproducible repository

Grading, assessment, and integrity

Grade policy (max 20)

Core: 0–13 pts + Extensions: 0–7 pts

- Up to 13/20 is achievable with Core tasks only.
- To exceed 13/20, integrate ≥ 1 Extension module into the main protocol and report narrative (not an isolated appendix).

Individual assessment

- Base grade is assigned to team deliverables (methodology, evidence, reproducibility).
- Individual grades may be adjusted based on oral discussion and self/peer assessment (contribution + understanding).
- Peer review: each student submits a short self/peer assessment.

— Academic integrity & AI tools

- All code, analyses, figures, and report text must be produced by the team; cite any external material.
- Generative-AI tools are allowed for consultation only (concepts/syntax/methodological alternatives); any consultation must be disclosed (tool + purpose) in the report or repository.
- AI-generated code, text, analyses, or experimental results are prohibited and treated as misconduct.
- Teams must be able to explain and reproduce any part of the work during the oral discussion.